

Stacked Solar-Shade Radiator Architecture

A Photovoltaic Canopy for Equatorial-Belt Lunar Surface Assets — Rev B

Architecture Reference Document | Rev B — September 2026 | Not for Hardware Commitment

1. Summary

A stacked solar-shade radiator places a photovoltaic canopy above a horizontal flat-plate radiator so that the canopy shades the radiator from the sun while generating power. Rev B rebuilds the February 2026 analysis with the radiator-to-canopy view factor computed for the actual geometry rather than assumed. A canopy large enough to shade the radiator occupies 55–99 % of its view of the sky, depending on height, and that narrows where the configuration pays: it is a net thermal loss at every sun angle for a clean radiator, and a clear gain for dust-degraded radiators under a tall canopy (3–4 m) at sites within about 15° of the equator. It is a fixed-asset configuration for low latitudes, not a vehicle roof, and polar sites do not need it.

What the architecture is, honestly stated: a solar canopy for fixed equatorial-belt assets that must run through lunar noon, valued for dust-robust rejection, for power from the same structure, and for a shaded work area beneath it. Every number here comes from a single closed-form model (`models/sssra_model.py`) whose only inputs are coating properties and geometry.

2. The problem

In vacuum every watt leaves by radiation. A horizontal radiator with a clear view of space rejects $\epsilon \sigma T^4 = 461 \text{ W/m}^2$ at 310 K with a clean white coating (ϵ 0.88), and absorbs $\alpha_s \cdot S \cdot \sin \alpha$ from the sun, where $S = 1,361 \text{ W/m}^2$ and α is the sun's elevation. The absorbed load is small at low sun and large at high sun:

Sun elevation	Incident on a horizontal surface (W/m²)	Absorbed, clean α_s 0.12	Absorbed, dusty α_s 0.25
6°	142	17	36
15°	352	42	88
30°	680	82	170
45°	962	115	241
60°	1,179	141	295
90°	1,361	163	340

At a polar site the sun grazes a horizontal radiator and the parasitic load is a few percent of the rejection. At an equatorial site at noon a dust-degraded radiator absorbs most of what it can emit. Shading is therefore a high-sun problem. The question the architecture has to answer is whether a shade that also blocks part of the sky is a net gain, and that depends on the view factor from the radiator to the shade.

3. Model

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q_unshaded =  $\epsilon_{\text{rad}} \sigma T_{\text{rad}}^4 - \alpha_s S \sin \alpha$ 
q_shaded   =  $\epsilon_{\text{rad}} \sigma T_{\text{rad}}^4 (1 - F) - F \epsilon_{\text{back}} \sigma T_{\text{array}}^4$ 
overhang (each sun-facing edge) =  $h / \tan \alpha_{\text{shade}}$        $F = F(\text{radiator}, \text{canopy}, h)$ 
computed numerically

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F is the fraction of the radiator's hemisphere occupied by the canopy underside. It is computed by direct integration over both surfaces for the actual rectangles; for equal parallel plates the closed form agrees to three figures (below). The canopy's underside radiates back onto the radiator at its own temperature; a low-emissivity backside coating (vacuum-deposited aluminium, $\epsilon_{\text{back}} \leq 0.1$) keeps that term small but does nothing about the sky it blocks. Reference properties: radiator ϵ 0.88 clean / 0.75 dusty, α_s 0.12 / 0.25, T_{rad} 310 K; canopy backside 330 K nominal, up to 370–400 K at equatorial noon.

View factor versus canopy height, 2 × 3 m radiator

Canopy height h	Same-size canopy, closed form	Same-size, numeric	Canopy with 1 m overhang all round	2 m overhang
0.4 m	0.73	0.73	0.96	0.99
0.7 m	0.59	0.59	0.90	0.96
1.0 m	0.48	0.48	0.82	0.92
1.5 m	0.34	0.34	0.68	0.83
2.0 m	0.25	0.25	0.56	0.74
3.0 m	0.15	0.15	0.38	0.57

The February document's 0.25–0.40 corresponds to a same-size canopy 2–3 m up. A same-size canopy cannot shade the radiator except with the sun overhead; a canopy that shades at moderate elevations needs overhang, and overhang fills the sky. The overhang needed on each sun-facing edge:

Canopy height	Shade at $\geq 6^\circ$	$\geq 15^\circ$	$\geq 30^\circ$	$\geq 45^\circ$	$\geq 60^\circ$	$\geq 75^\circ$
0.4 m	3.8 m	1.5 m	0.7 m	0.4 m	0.2 m	0.1 m
0.7 m	6.7 m	2.6 m	1.2 m	0.7 m	0.4 m	0.2 m
1.0 m	9.5 m	3.7 m	1.7 m	1.0 m	0.6 m	0.3 m
1.5 m	14.3 m	5.6 m	2.6 m	1.5 m	0.9 m	0.4 m
2.0 m	19.0 m	7.5 m	3.5 m	2.0 m	1.2 m	0.5 m
3.0 m	28.5 m	11.2 m	5.2 m	3.0 m	1.7 m	0.8 m

4. Net rejection, shaded against unshaded

Net rejection in W/m^2 of radiator. Shaded columns are for a canopy with 1 m of overhang all round and a coated backside at 330 K; the shaded figure does not depend on sun angle by construction. Unshaded figures fall with sun elevation.

Sun	Unshaded clean	Unshaded dusty	Canopy 0.4 m, clean / dusty	Canopy 0.7 m, clean / dusty	Canopy 1.0 m, clean / dusty	Canopy 1.5 m, clean / dusty	Canopy 2.0 m, clean / dusty	Canopy 3.0 m, clean / dusty
2°	455	381	-47 / -50	-13 / -20	30 / 17	103 / 81	167 / 137	262 / 220
6°	444	357	-47 / -50	-13 / -20	30 / 17	103 / 81	167 / 137	262 / 220
10°	432	334	-47 / -50	-13 / -20	30 / 17	103 / 81	167 / 137	262 / 220
15°	419	305	-47 / -50	-13 / -20	30 / 17	103 / 81	167 / 137	262 / 220
20°	405	276	-47 / -50	-13 / -20	30 / 17	103 / 81	167 / 137	262 / 220
30°	379	223	-47 / -50	-13 / -20	30 / 17	103 / 81	167 / 137	262 / 220
45°	345	152	-47 / -50	-13 / -20	30 / 17	103 / 81	167 / 137	262 / 220
60°	319	98	-47 / -50	-13 / -20	30 / 17	103 / 81	167 / 137	262 / 220
75°	303	64	-47 / -50	-13 / -20	30 / 17	103 / 81	167 / 137	262 / 220
90°	297	52	-47 / -50	-13 / -20	30 / 17	103 / 81	167 / 137	262 / 220

Break-even: the lowest sun elevation at which the shaded radiator out-rejects the unshaded one.

Canopy height	F	Shaded clean / dusty (W/m²)	Beats unshaded clean from	Beats unshaded dusty from
0.4 m	0.96	-47 / -50	never	never
0.7 m	0.90	-13 / -20	never	never
1.0 m	0.82	30 / 17	never	never
1.5 m	0.68	103 / 81	never	67°
2.0 m	0.56	167 / 137	never	49°
3.0 m	0.38	262 / 220	never	31°

A clean white radiator is never improved by a shade that blocks its sky, because it absorbs at most a quarter of the sunlight it is shaded from while losing the whole of the sky the canopy covers. The case for the canopy is the dust-degraded radiator at high sun, where absorption has doubled and emission has fallen, and only under a canopy tall enough that its view factor drops below about 0.5.

5. Where it pays: site latitude

For a fixed asset the sizing case is local noon, when the sun reaches 90° minus the latitude. The canopy is sized to shade the radiator for elevations within 20° of that maximum (about 79 hours either side of noon at the Moon's 0.5°/h); outside that band the sun is lower and the unshaded load falls with it. Shaded figures are clean / dusty, canopy backside 330 K.

Site latitude	Noon elevation	Unshaded at noon, clean / dusty	Canopy 1.0 m: F, shaded, canopy size	Canopy 2.0 m: F, shaded, canopy size	Canopy 3.0 m: F, shaded, canopy size	Canopy 4.0 m: F, shaded, canopy size
0°	90°	297 / 52	F 0.65: 120 / 96 W/m ² , 2.7 × 3.7 m	F 0.48: 205 / 170 W/m ² , 3.5 × 4.5 m	F 0.40: 251 / 210 W/m ² , 4.2 × 5.2 m	F 0.34: 280 / 235 W/m ² , 4.9 × 5.9 m
15°	75°	303 / 64	F 0.75: 63 / 46 W/m ² , 3.4 × 4.4 m	F 0.64: 121 / 97 W/m ² , 4.8 × 5.8 m	F 0.58: 153 / 125 W/m ² , 6.2 × 7.2 m	F 0.54: 173 / 142 W/m ² , 7.6 × 8.6 m
30°	60°	319 / 98	F 0.84: 15 / 4 W/m ² , 4.4 × 5.4 m	F 0.78: 47 / 32 W/m ² , 6.8 × 7.8 m	F 0.75: 65 / 48 W/m ² , 9.2 × 10.2 m	F 0.73: 76 / 58 W/m ² , 11.5 × 12.5 m
45°	45°	345 / 152	F 0.92: -27 / -32 W/m ² , 6.3 × 7.3 m	F 0.90: -14 / -21 W/m ² , 10.6 × 11.6 m	F 0.89: -7 / -15 W/m ² , 14.9 × 15.9 m	F 0.88: -3 / -12 W/m ² , 19.2 × 20.2 m
60°	30°	379 / 223	F 0.98: -58 / -59 W/m ² , 13.3 × 14.3 m	F 0.98: -56 / -58 W/m ² , 24.7 × 25.7 m	F 0.98: -56 / -57 W/m ² , 36.0 × 37.0 m	F 0.98: -55 / -57 W/m ² , 47.4 × 48.4 m
85°	5°	447 / 363	F 0.99: -64 / -65 W/m ² , 24.9 × 25.9 m	F 0.99: -64 / -65 W/m ² , 47.7 × 48.7 m	F 0.99: -64 / -65 W/m ² , 70.6 × 71.6 m	F 0.99: -64 / -64 W/m ² , 93.4 × 94.4 m

Reading the table: at the equator a 3 m canopy holds a dusty radiator at 210 W/m² where an unshaded one has fallen to 52, a four-fold difference, while giving up 46 W/m² when clean. At 15° latitude the dusty advantage is still two-fold. At 30° the overhang has grown to fill three quarters of the sky and the canopy loses in every state; beyond that the numbers are negative, meaning the canopy underside heats the radiator more than the radiator can shed past it. The architecture's region is the equatorial belt, roughly ±15° of latitude, with a canopy 3 m or more above the radiator.

Sensitivity to canopy temperature and backside coating, equatorial design point

Canopy 3 m up, shading from 70°, F = 0.40. Dusty radiator, net rejection in W/m²:

Canopy backside temperature	Coated backside ϵ 0.1	Bare backside ϵ 0.82
300 K	219	88
330 K	210	18
370 K	195	-108
400 K	179	-235

The low-emissivity backside is necessary: without it the canopy is a heater at any realistic temperature. It is not sufficient; the sky it blocks is the dominant term at every height.

6. Sizing for an equatorial installation

Canopy 3 m above the radiator, shading guaranteed for elevations of 70° and above (overhang 1.1 m on each edge), dusty end-of-mission properties for the radiator area, square radiator for the estimate. Radiator mass 8–12 kg/m² with fluid loop; canopy 4–8 kg/m² of array plus truss at 1.5 kg/m² per metre of height and four posts.

Heat load	Radiator area, clean	Radiator area, dusty (sizing)	Radiator side	Canopy footprint	Canopy power at noon	Radiator mass	Canopy + truss mass
500 W	2.0 m²	2.4 m²	1.5 m	14 m²	5.3 kW	19–29 kg	158–213 kg
1000 W	4.0 m²	4.8 m²	2.2 m	19 m²	7.3 kW	38–57 kg	202–278 kg
2000 W	8.0 m²	9.5 m²	3.1 m	28 m²	10.6 kW	76–114 kg	276–387 kg
5000 W	19.9 m²	23.8 m²	4.9 m	50 m²	19.0 kW	190–285 kg	464–663 kg
10000 W	39.8 m²	47.5 m²	6.9 m	82 m²	31.4 kW	380–570 kg	741–1070 kg

The canopy is much larger than the radiator it protects, which is the point for a fixed asset: the excess area is power and shaded ground. The same fact rules the configuration out for anything that has to move.

7. What the polar program uses instead

Aegis Station Infrastructure's surface program is south-polar, where the thermal and power problems separate. At 6° sun a horizontal surface receives 142 W/m² and a clean white radiator absorbs 17 W/m² of it; an unshaded roof radiator rejects 444 W/m² clean and 357 W/m² dusty. The Aegis-Class Rover therefore carries a 6 m² unshaded roof radiator (2,143 W dusty, 2,662 W clean) and no roof array: a

horizontal 20 m² array at 6° makes 797 W, the same area held normal to the sun makes 7,622 W, and a same-size array 0.4 m above the radiator would have cut its rejection to 74 W/m². The rover's power comes from deployable sun-tracking wings when parked; a tracking mast array for driving is an open item. The only element of the February concept retained on the polar vehicles is the roof plate that carries the node power and thermal connection, an ASI interface standard shared by the rover and the Short Hopper. It does not depend on the canopy.

Polar power is its own problem and is not solved by anything on a roof. With the sun circling the horizon, the productive array is vertical and slowly rotating; NASA is currently soliciting vertical deployable solar arrays for the south pole for exactly this reason. For a base rather than a vehicle, the likely mix is vertical arrays on persistently lit high ground, batteries for the shadow periods, fission surface power, and possibly transmission lines from the lit ridges down to the sites that need the power. Those are the inputs the Aegis surface architecture plans around; the canopy is not one of them.

8. Dust

Dust raises the radiator's solar absorptance and lowers its emissivity; the reference properties above assume a 21-day exposure. The canopy removes the absorptance term entirely, which is why its advantage appears only in the dusty columns, but it does nothing for the emissivity loss, and the canopy's own top surface degrades the same way as any array. The canopy does keep settling dust and ballistic ejecta from above off the radiator face. Electrodynamic dust shields, periodic tilting and simple covers apply to both surfaces and are not specific to this architecture.

9. Applicability

Platform	Sun regime	Verdict
Equatorial-belt surface habitat or ISRU plant (fixed, ±15° latitude)	noon elevation 75–90°	Canopy 3–4 m: dust-robust rejection, power, shaded work area. Recommended where the asset must run through noon.
Logistics node in the equatorial belt	as above	As above; the canopy doubles as the node's array.
Mid-latitude fixed asset (30–60°)	noon elevation 30–60°	Not recommended; the overhang needed fills the sky. Use an unshaded radiator sized to dusty properties, or a parasol that does not cover the radiator's zenith view.
Polar asset, fixed or mobile	≤ 15°	Not applicable; an unshaded horizontal radiator is close to ideal.
Any mobile platform	any	Not recommended; a 3 m canopy is not a vehicle roof, and a 0.4 m roof stack cuts rejection by three quarters.
Permanently shadowed region	none	Not applicable.

10. Conclusions and recommendations

1. Compute the view factor for the real geometry. Rev A assumed 0.25–0.40; the real range for a shading canopy is 0.4–0.99, and it governs.
2. Treat the stacked solar-shade radiator as an equatorial-belt canopy for fixed assets, sized for noon, 3 m or more above the radiator, with a coated backside. Do not treat it as a general lunar thermal architecture.
3. Do not put it on a vehicle. Rovers carry an unshaded radiator sized to dusty properties and a tracking array.
4. Keep the roof power-and-thermal interface plate as the ASI standard; it is independent of the canopy.
5. Before any hardware commitment at an equatorial site, replace the constant-temperature canopy assumption with a coupled canopy thermal model and validate the coated-backside emissivity after dust exposure.

Rev B, September 2026. Replaces the February 2026 reference. Model and tables:
~/sssra/models/sssra_model.py.
